OPINION

The Radiologist's Imitation Game

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The recently released films *The Imitation Game*, *Her*, and *Ex Machina* have brought a philosophical quandary to the forefront of popular culture: can machines think or, at least, display behavior indistinguishable from that of a thinking human? Although it may seem silly to even entertain this question, the philosophical issues raised by the attempts to answer it offer important insights into some of the most pressing problems currently facing radiology: de-skilling, automation, and a sense of being pushed to the periphery of medicine in the era of ever more centralized health care administration and reimbursement.

"Can machines think?" was initially asked by the founder of modern philosophy, Rene Descartes, in his 1637 *Discourse on Method*. Descartes, who knew of the relatively crude automatons of the 17th century, was compelled by his peculiar metaphysics to answer in the negative [1]. This question has been taken up several times in the intervening centuries, most famously by Alan Turing, the British mathematician, cryptographer, and computer scientist. Turing's test, which provides the title for the biopic *The Imitation Game* and the plot for *Ex Machina*, can be summarized as follows: Imagine that one person (the interrogator) is given the task of asking yes-or-no questions of an anonymous interlocutor she cannot see or talk to and who responds to her questions via typewritten answers. By asking questions she thinks only a human could answer, the interrogator must determine if her anonymous interlocutor is human or a computer. If a computer could fool the interrogator into thinking it is human, it has passed the Turing test [2].

In the 1960s and 1970s, some computer scientists suggested that a machine running software that could pass the Turing test would not merely "think" in some qualified sense of the word but would think in exactly the same way human beings think. A truly conscious mind would exist in the computer-software combination. This position is often called "strong artificial intelligence."

In response to the strong artificial intelligence position, philosopher John Searle devised a thought experiment known as "the Chinese room" [3]. Imagine a man who speaks English and no Chinese, locked in a room with pens, paper, and a set of rules for manipulating Chinese characters. A set of Chinese characters would be given to him through a slot in the door, he would follow the rules, and he would produce another set of Chinese characters, which he would send out of the room. Unbeknownst to the man in the room, the initial characters would constitute a question and the set of characters he produced would be an answer.

The man in the room is an analogy for the central processing unit of a computer, the list of rules is the software, and the pens and paper are the hardware. Although "the room" seems to "know" Chinese, the person in the room would have no idea what the questions were or what answers he was giving. Searle's point isn't that machines can't think (he thinks the opposite, that humans are a special kind of thinking machine) but rather that appropriate manipulation of symbols according to formal rules does not constitute thought. Thought requires understanding the meaning of the symbols we are using and how these symbols ultimately relate to the wider world.

There is debate as to whether the Chinese room experiment proves Searle's point about how the mind and computers work. But it does offer an illustration of a less controversial phenomenon: a real flesh-and-blood human engaged in an activity that looks like thinking but is in fact not thinking.

The situation of the radiologist is analogous to the Chinese room. CT scans and radiographs come in; reports flow out. Is the person in the room thinking? Or is he following a set of rules?

As a resident on call, I ought to be thinking and aware of the fact that radiographs and CT scans refer to real people and that clinicians are depending on my interpretations to make serious medical decisions. However, on busy nights, I have found myself, from time to time,

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looking into clinical and laboratory data less than I should and trying to dictate as quickly as possible. Head CT studies become occasions to “look for bright stuff in the brain and make sure that the bunny ears are dark and symmetrical,” and pulmonary embolism protocol CT studies become occasions to “look for dark stuff in those round things that are supposed to be white.” All the while, I’m producing dictations that contain phrases such as “no intra- or extra-axial fluid collections,” “no midline shift,” and “no pulmonary embolism.” While lapsing into this mode of semiconsciousness, I’ve even thought to myself, “I’m a machine.”

These brief moments of mindlessness have not caused any major problems for me or my patients, but occasionally an attending physician will notice a problem with my differential diagnosis or a subtle missed finding, which is a direct result of treating radiology as if it is a game of symbol manipulation. Appropriately, that physician will ask me at readout, “What were you thinking?”

As a resident, I am far more prone to slip into this kind of rote pattern recognition than more experienced radiologists. There are situational factors at work as well; being tired and hungry isn’t conducive to deep thought. Moreover, even if I wanted to get more clinical information, often none is available. I’m forced to apply abstract rules to generate a differential diagnosis of dubious relevance, rather than to integrate history, physical examination findings, laboratory values, and prior imaging into something more coherent, and, hopefully, more relevant. Even when I am thinking, my work can feel disconnected from the real flesh-and-blood patients who depend on my interpretations.

Matthew Crawford, philosopher and motorcycle mechanic, chronicled the ways in which work has been de-skilled during the 20th century in *Shop Class as Soulcraft* [4]. At first, this de-skilling affected only manual laborers, such as the early 20th century craftsmen who hand-built cars and were replaced by Henry Ford’s assembly-line workers. However, with the growth in computing power, jobs that once required a human with a fair bit of knowledge, such as preparing taxes, can now be done by a computer. Theoretically, with enough computing power, any job that involves manipulating symbols by following explicit rules could be performed by a computer, a robot, or a non-thinking person, like the translator in the Chinese room. Only jobs that require tacit knowledge, knowledge that cannot be expressed as abstract symbols, will be safe from de-skilling, routinization, and automatization.

Crawford also reminds us of an old distinction between knowing facts about the world and knowing how to do things within the world. Because of the success of automatization in heavy industry, the US educational system has emphasized knowing facts as the path to economic security. But Crawford asks his readers to reconsider the supremacy of knowing facts over knowing how to do things. For Crawford, not only is knowing how to do something somewhat protective from the pressures of the modern economy, but it is a distinctive moral act. To learn any skill, we are forced to engage with the world, submit to its ways, and endure the frustrations and failure that such engagement often entails. This process changes people for the better and rewards them with the knowledge that they have made a tangible difference in the world. He cites mechanics, electricians, plumbers, musical instrument makers, and doctors as estimable examples of people who have struggled to acquire real skills and who can consequently make real differences in the world.

Admittedly, much of the appeal of radiology lies in the fact that radiologists hover above the clinical fray, a bit like disembodied minds in one of Descartes’s famous thought experiments. Rarely do we encounter “code brown,” Fournier’s gangrene, or the challenges of a difficult hospital disposition. Reading rooms are generally comfortable places where coffee can be consumed. Yet the best radiologists, the ones whom I hope to be like, eventually, are ones who are willing to leave the reading room when the situation calls for it. Moreover, they have a deep knowledge of the craft of medicine. Some have completed other residencies before radiology. Others know the details of radiographic and MRI coil positioning or how to put a feeding tube in the most difficult of patients. Still others have incredibly detailed understanding of surgical procedures and instruments or of genetics and pathology. None of these expert radiologists would ever be mistaken for an automaton.

A perennial fear of radiologists, especially since the invention of PACS and the loss of physical control over film, has been replacement: by other clinicians, by physicians reading remotely over the Internet, or by computers. Economic, professional, technical, and legal constraints have prevented this from happening. Yet unease about the future remains. The good news is that radiologists seem to have diagnosed the problem. It seems as if every week I get an e-mail or hear a lecture about the need for radiologists to be more collaborative and engage with clinicians and patients. However, the suggestions always seem a bit vague.

A study presented at the RSNA meeting in 2008 found that by allowing interpreting radiologists to view patients' faces, they were more likely to feel empathy toward the patients and, perhaps more important, were more methodical in their image interpretation, with fewer missed incidental findings [5]. A recent editorial in *JACR* documented the success of one radiology group stationing a radiologist in the doctor's lounge [6]. Although concerns for patient privacy and physician productivity have prevented these practices from being widely adopted, the results point to something deep in human nature. It is through our faces that we reveal ourselves to the world.

As a radiology resident, this is clear. During a face-to-face readout, mistakes and missed findings take on a personal dimension they would never have if readout were conveyed to me in an electronic typewritten form. Readout should not be like a Turing test, between two anonymous individuals, but rather a conversation between two people who inhabit a shared profession. Perhaps, if we can't actually station ourselves in the doctor's lounge, radiologists ought to attach our pictures to our reports. It would be a small step in becoming more real to our patients and to consulting physicians who would otherwise never know us.

In the film *Ex Machina*, the Turing test is modified. The computer being tested has a humanoid body and, more important, a beautiful face. Without giving too much away, this complicates the test, as it introduces an emotional component into what was initially conceived as a test of disembodied intelligence. The human face has been a key feature of our literary, artistic, religious, and philosophical heritage [7]. Think of the self-portraits of Rembrandt or Steve McCurry's famous *National Geographic* photograph "The Afghan Girl." No one can doubt that the people in these images have rich interior lives. The essence of the interior life is thought and understanding, not the manipulation of formal symbols according to an arbitrary syntax.

One of the key features of both the Turing test and the Chinese room is that face-to-face human interaction is removed. The continued popularity of these thought experiments points to the fact that in our culture we have accepted a tacit dualism, and we conceive of thinking as the purview of a disembodied mind. It seems to me that the future of radiology depends on our willingness to abandon this line of thinking and to embrace a more corporeal understanding of our profession. This may mean doing something as simple as attaching our faces to our reports or seeing (and smelling) some things we would rather not.

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